

ZapBox Simple – Operating Instructions

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Overview

The **ZapBox Compact** is an electronic switch for Bitcoin Lightning payments. A payment via the Lightning Network can be used to switch an output – ideal for vending machines, presentations, event control, and many other applications.

Basic Equipment

Component	Description
Microcontroller	T-Display-S3 with 1.9" LCD display
Front panel	35° display
Input	Dual USB-A and USB-C
Output	Dual USB-A and USB-C
Control	Two on-board micro switches
Option BTC ticker	Activatable via the Web Installer

Views

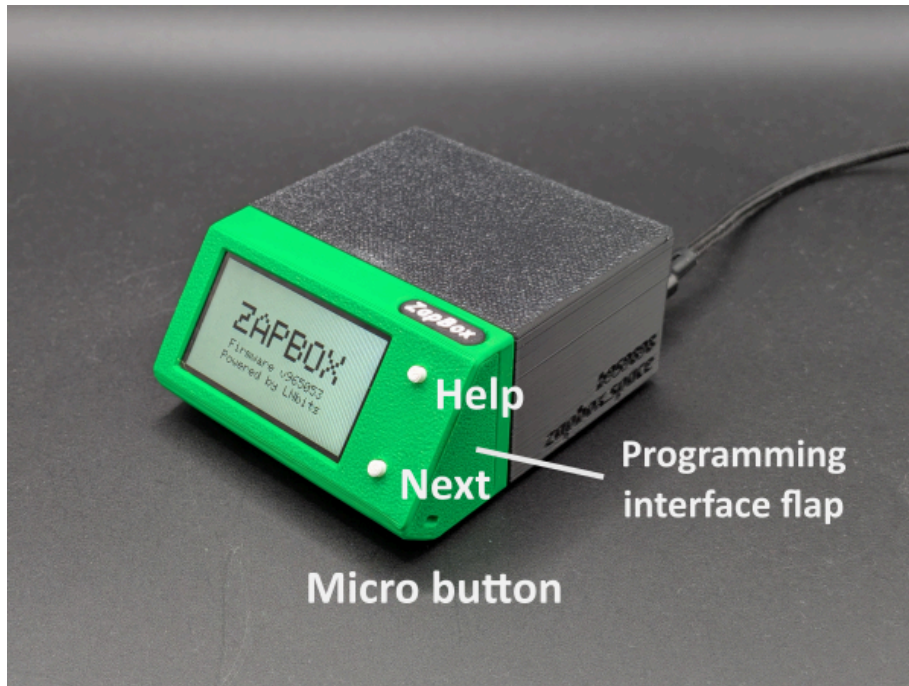


Figure 1: Front view



Figure 2: Rear view

Connections

Input - Dual USB-A and USB-C socket for power supply (5V)

Power the device via the **Power IN** connector using a USB-C/A cable with **5 V DC (max. 5 A)**.

Note: The USB power connector does not support automatic USB-C power negotiation (no USB-C Power Delivery). Some USB-C chargers or power modules therefore do not recognize the ZapBox as a load and will not supply power. In this case, use a **USB-A output** of the power supply or an alternative 5 V power source. The maximum current must not exceed 3 A.

Input - USB-C socket on the microcontroller (data access, behind the right side panel)

To read or transfer data from the device, connect the ZapBox to a computer or laptop:

1. On the **right side of the front panel** there is a small, concealed flap.
2. Open the flap by pushing it to the right from below using a **narrow screwdriver**.
3. Connect a USB-C cable to the microcontroller's connector underneath.



Figure: Opening the panel and USB-C connector for data

Important note: The USB connector directly on the microcontroller is intended exclusively for flashing the firmware and transferring configuration parameters. During the flashing process, no load may be connected or switched at the output, as this can lead to malfunctions or **damage to the microcontroller**.

It is therefore recommended to either:

- not connect any load at the output while the USB connection is active, or
- additionally connect the regular **Power-IN input** to the same power supply. This ensures that the current for the power relay does not flow through the microcontroller and overload it.

Error Diagnosis and Troubleshooting

The ZapBox has a convenient error display via the screen. There are four basic errors, which are prioritized:

Prio.	Error type	Abbreviation	Detection method	Description
1	NO WIFI	NW	WiFi connection status	WiFi network not connected -> Is the WiFi data correct? -> Is the WiFi signal too weak?
2	NO INTERNET	NI	HTTP check to Google	Internet connection lost -> Is the internet reachable?

Prio.	Error type	Abbreviation	Detection method	Description
3	NO SERVER	NS	TCP port 443 check	LNbits server unreachable -> Has the server hardware failed? -> Is the device string correct?
4	NO WEBSOCKET	NWS	WebSocket connection status	WebSocket protocol/handshake error -> Has LNbits failed? -> Is the device string correct?

The error messages are also logged and can be retrieved via the *Report Mode*:

- Press the HELP button twice quickly in succession to show error counters (0-99) for all four error types with their occurrence frequencies.
- Press the LED button three times quickly in succession (if an external LED button is available).

Further up-to-date information on error descriptions can be found on the Web Installer page in the "Error Detection & Report" and "Troubleshoot" chapters.

Output - Dual USB-A and USB-C (switched 5V voltage)

The USB sockets are switched via a relay contact. The **total load** of the sockets should **not exceed 3 A**.

Controls

The ZapBox has two small **on-board micro switches** that are connected directly to the microcontroller. All functions can be accessed via the micro switches. In addition, the ZapBox has a reset button on the underside of the front panel and two slide switches on the side.

Function Overview - Micro Switches

Function	Micro switch
Show help page	Press HELP once
Next page / product change	Press NEXT once
Show REPORT page	Press HELP twice
Open config mode	Hold NEXT for at least 5 sec.

Setup and Commissioning

The ZapBox is tested after manufacturing and shipped with the current firmware - however, it is not yet parameterized. The software is under active development, so it is recommended to flash the ZapBox with the latest firmware right at the start and then perform the parameterization. A convenient **Web Installer** is available for this.

Step 1: Firmware Update

1. Open the right side panel of the front panel, as described above under "Input - USB-C socket on the microcontroller".
2. Connect the ZapBox to the USB-C port with a cable and connect it to a computer.
3. Open a Chromium browser, for example Google Chrome, Microsoft Edge, Brave, Vivaldi, Opera, or [Helium](#).

Step 2: Parameterization

1. Navigate to the Web Installer page in the browser.
2. Follow the instructions on the page to enter the desired parameters such as **WiFi SSID**, **WiFi password**, and **Device Settings String**.
3. Save the settings and restart the ZapBox.

Note: No load should be connected to the outputs during setup, to avoid malfunctions or damage to the microcontroller.

After initialization, the ZapBox will display the product's QR code and is ready for the first payment and subsequent switching action.

Technical Data

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	5.0 A
Output power	max. 3.0 A (recommended)
Display	1.9" LCD (T-Display-S3)
Temperature range	0–40 °C
Communication	Wi-Fi (ESP32-S3)
Payment protocol	Bitcoin Lightning Network

Safety Instructions

- Operate the device only with the specified supply voltage.
 - Do not exceed the maximum current load of the outputs.
 - Do not perform any work on the relay contacts under load.
 - The device is not suitable for use in damp or wet environments.
 - Ensure adequate ventilation around the device.
 - Keep out of the reach of children.
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Further Links

Resource	Link
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick overview & troubleshooting	https://installer.zapbox.space/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files, operating instructions, etc.)	https://github.com/AxelHamburch/ZapBox
ZapBox Extension	https://github.com/AxelHamburch/zapbox_extension
LNbits	https://lnbits.com/

Subject to changes and errors. As of: 2026